

ADR 0150 — The live graph follows history generation, not the planner token

- **Status:** Proposed — implemented for #852
- **Date:** 2026-09-18
- **Issue:** [#852](#)
- **Extends:** [0094](#) (the change feed is a reading, never a graph invalidation) · [0001](#) (distinct generation recipes) · [0115](#)
- **Supersedes:** nothing
- **Amends:** [0094](#) § what the feed is *for* — it still does not rebuild a plan, and it still does not compare its token to the graph's

Context

M12 taught the app to *say* when the repository moved underneath it. The change feed publishes the planner generation. The plan-freshness panel reads that feed. The live commit graph did not.

So a `git commit` in a terminal, an auto-checkpoint, or a checkout left the canvas showing the previous history until the user hit Refresh. In-app writes already remounted through `GraphCore::on_invalidate`. External writes only changed the freshness badge.

The obvious wiring — `on_invalidate` with the feed's generation — is the mix `staging.rs` already names: it "409s forever, never admits". Five generation recipes ship in this server. The feed carries the planner recipe (HEAD, refs, worktree status). The graph is pinned to history-v1 (committed topology, plus shallow). Those tokens are not comparable. Treating them as one would

remount the canvas on every editor save, and after an in-app write it would remount *again* because the stored `GraphCore` token from settlement is also the planner recipe.

Change feed

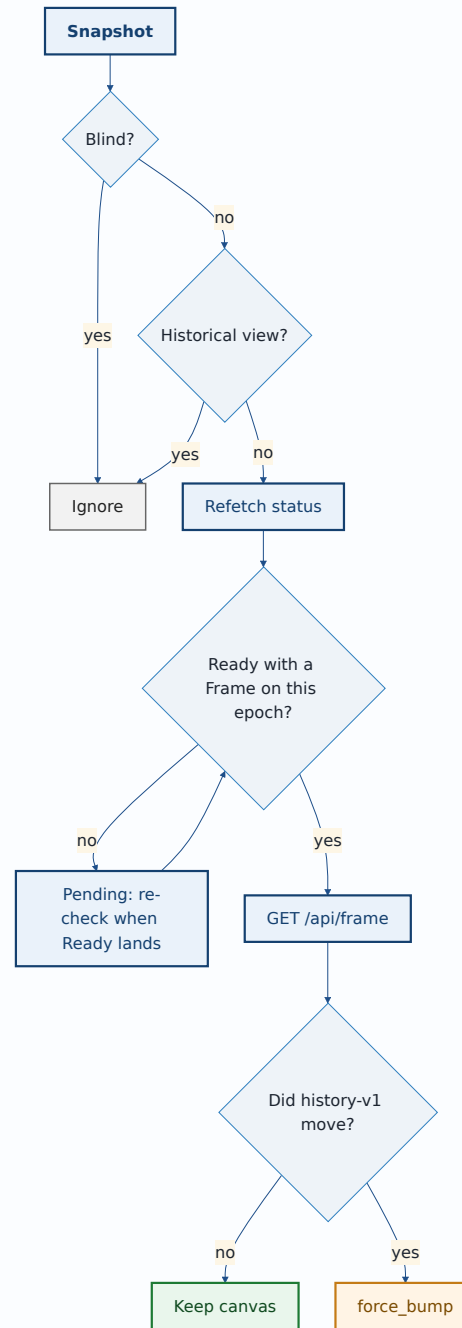
Live graph

**Do not
compare their
tokens**

Decision

The feed remains a hint. History-v1 remains the fact.

1. A change-feed snapshot with a generation is a **reading**. Blind (`generation: None`) is not. The first snapshot on a stream, and every snapshot after a gap, still count: `RefDelta::Unknown` is the absence of a named delta, not the absence of a reading.
2. Every reading refetches working-tree status. The chip is allowed to move when only the worktree moved.
3. The canvas remounts only when a live `GET /api/frame` disagrees with the Frame already on this epoch. That comparison is history-v1 against history-v1.
4. No displayed Frame (seed still loading) does **not** remount. Bumping then fights the in-flight seed. The Ready transition re-checks, so an external commit during first load cannot strand a stale page.
5. A historical as-of view does not follow live history. Refresh in that mode is still "reload this observation", never "return to live".
5. `GraphCore::on_invalidate` is not this path. After a write it stores the planner token. Feeding it a history-v1 token would look like movement every time.



A late Frame for a retired epoch is dropped: the probe captures the epoch, awaits the Frame, then re-reads the live epoch before bumping.

Alternatives considered

Alternative	Why it lost
<code>on_invalidate</code> with the feed token	Mixes planner and history-v1. Remounts on editor saves. After an in-app write, remounts a second time.
Remount on any <code>RefDelta::Named</code> with refs	HEAD's symbolic target is folded into <code>other</code> with worktree status. A checkout would not move the HEAD badge.
Skip <code>other</code> entirely	Same miss: checkout is <code>other</code> .
Put history-v1 on the change feed	A protocol change for a client that can already read <code>/api/frame</code> . The over-read of one extra Frame per repository change is the cheaper mistake.
Leave the graph until Refresh	That is the bug. The feed already knows something moved; keeping a picture of a repository that no longer exists is the state M12 removed for plans and then left standing for the graph.

Consequences

External commits, checkouts, and ref movement reload the live graph without a manual Refresh. Worktree-only edits refresh the status chip and leave the camera alone.

In-app writes still remount once through settlement. The follow-up probe then sees the same history-v1 Frame and does not bump again, provided the seed for that epoch has landed. A snapshot during `SeedLoading` waits for Ready.

The wasm wrapper is not compiled by `cargo test`. The decisions live in `features/freshness/core.rs`. A source census pins that `signals.rs` asks those functions, probes `/api/frame`, skips `on_invalidate`, and re-reads the epoch after the await.

Failure-atlas **678** (history comparison inverted) and **679** (reload with no displayed Frame, which would fight the in-flight seed) are both conclusive `caught`, at disjoint assertions.

Signed: grok · 2026-09-18T05:25:00Z